Calculus

Integration by substitution

8/19/2026 Lecture

Substitution

Substitution used for finding the anti-derivatives of complex functions.
Let's look some examples.

Some functions couldn't be integrated directly. For instance,

$$f(x) = \frac{1}{\sqrt{6 - x^2}}$$

We can't evaluate the anti-derivative of this directly.
In another case, there's a function

$$f(x) = \frac{1}{1 + e^x}$$

WTF.

Both looks completely not integrable, but it integrated by substitute new variables corresponding to current variables.

Substitution – For single variable

Let's look at the function below,

$$f(x) = \frac{1}{1 + x}$$

And anti-derivatives of function must be:

$$F(x) = \int \frac{1}{1 + x} dx$$

1+x is in denominator of fraction, it couldn't integrate directly by x.
If we simply define 1+x to new variable, like below,

$$let\ u = 1 + x \rightarrow \frac{dx}{du} = 1$$

dx becomes du, then we can re-write

$$F(x) = \int \frac{1}{1 + x} dx = \int \frac{1}{u} du = \ln|u| + C = \ln|1 + x| + C$$

Techniques – u-method

To integrate complex fractional functions have form like below,

$$\frac{x^{n-1}}{a+x^n}$$

$$let\ a+x^n=u \rightarrow \frac{du}{dx}=\ nx^{n-1}$$

Exercises 1:

Find anti-derivatives of given function:

$$f(x)=\ \frac{x}{2+x^2}$$

Exercises 2:

Find anti-derivatives of given function by substitution:

$$f(x)=\frac{4+x^2}{3+x^2}$$

Exercises 3:

Find anti-derivatives of given function:

$$f(x)=\frac{1}{x(x^6+2)}$$

Exercises 4:

Find anti-derivatives of given function by substitution:

$$f(x)=\frac{e^x}{e^{2x}+3e^x+2}$$

Substitution – Function has a sqrt

Now look at the function that has root. It's one of the simplest.

Let's get a function that has square root like below:

$$f(x) = \sqrt{1+x}$$

And anti-derivatives of function must be:

$$F(x) = \int \sqrt{1+x}\, dx$$

1+x is in denominator of fraction, it couldn't integrate directly by x.
If we simply define 1+x to new variable, like below,

$$let\ u = 1 + x \rightarrow \frac{dx}{du} = 1$$

dx becomes du, then we can re-write

$$F(x) = \int \sqrt{1+x}\, dx = \int \sqrt{u} du = \frac{2}{3}u^{\frac{3}{2}} + C = \frac{2}{3}(1+x)^{\frac{3}{2}} + C$$

Techniques – Root function

To integrate complex root functions have form like below,

$$x^{n-1}\sqrt[i]{a+x^n}$$

$$let\ a + x^n = u \rightarrow \frac{du}{dx} = nx^{n-1}$$

Exercises 1:

Find anti-derivatives of given function:

$$f(x) = \sqrt{2 - x}$$

Exercises 2:

Find anti-derivatives of given function by substitution:

$$f(x) = x^2\sqrt{4 - x^3}$$

Exercises 3:

Find anti-derivatives of given function:

$$f(x) = \frac{\sqrt{x^4 + 1}}{x^3}$$

Exercises 4:

Find anti-derivatives of given function by substitution:

$$f(x) = \frac{1}{x \ln x \sqrt{4 + (\ln x)^3}}$$

Substitution – Trigonometric substitution

That's the trickiest one. (Also, the funny one)
Image that the functions have both above condition instantly.

Like the below,

$$f(x) = \frac{1}{\sqrt{4 - x^2}}$$

If letting $4+x^2 = u$,

$$\frac{du}{dx} = -2x \rightarrow dx = -\frac{du}{2x}$$

$$\int \frac{1}{\sqrt{4 - x^2}} dx = -\int \frac{1}{\sqrt{u}} \cdot \frac{du}{2x}$$

The extra x couldn't be canceled out!
Thus, to eliminate x, we need to account for other variables. Trigonometric functions are good instances for alternative.

Let x to some trigonometric function. Like below:

$$let\ x = 2\sin\theta$$

$$\frac{dx}{d\theta} = 2\cos\theta \rightarrow dx = 2\cos\theta\, d\theta$$

$$\therefore \int \frac{1}{\sqrt{4 - x^2}} dx = 2\int \frac{\cos\theta}{\sqrt{4 - (\sin\theta)^2}} d\theta = 2\int \frac{\cos\theta}{2\cos\theta} d\theta = \theta + C$$

$$x = 2\sin\theta \rightarrow \theta = \sin^{-1}\frac{x}{2}$$

$$Thus, \int \frac{1}{\sqrt{4 - x^2}} dx = \sin^{-1}\frac{x}{2}$$

It is simply expressed as an inverse function of sine.

Also, we able to use tangent, and secant for antiderivatives of complicated functions.

Techniques – Trigonometric substitution

To integrate complex functions, consisting of one of them below,

* Even without root signs (radicals), trigonometric substitution is highly effective whenever you encounter expressions that match or can be converted into the standard Pythagorean identity forms

$$\sqrt{a^2 - x^2} \rightarrow let\ x = a\sin x\,, at\ the\ range\ of\ -\frac{\pi}{2} \leq \theta \leq \frac{\pi}{2}$$

$$\sqrt{a^2 + x^2} \rightarrow let\ x = a\tan x\,, at\ the\ range\ of\ -\frac{\pi}{2} < \theta < \frac{\pi}{2}$$

$$\sqrt{x^2 - a^2} \rightarrow let\ x = a\sec x\,, at\ the\ range\ of\ 0 \leq \theta < \frac{\pi}{2}\ or\ \pi \leq \theta < \frac{3\pi}{2}$$

Exercises 1:

Find anti-derivatives of given function:

$$f(x) = \sqrt{4 - x^2}$$

Exercises 2:

Find anti-derivatives of given function by substitution:

$$f(x) = \frac{x^2}{\sqrt{x^2 - 3}}$$

Exercises 3:

Find anti-derivatives of given function via trigonometric substitution:

$$f(x) = (x^2 + 2x + 5)^{\frac{3}{2}}$$